# WEEK 6 – Power BI

# HEALTHCARE ANALYTICS DASHBOARD

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## Week 5 Overview

Week 5 focused on **healthcare dataset cleaning and data modeling** in Power BI. The dataset was cleaned, duplicates were removed, and tables such as **Dim_Patient, Dim_Admission, and Fact_Billing** were created. Relationships were established between the tables, and **DAX measures** were created for billing, patients, length of stay, and readmission rate.

## Introduction

In **Week 6**, a **Healthcare Analytics Dashboard** was developed using Power BI to analyse **patient admissions, billing amounts, medical conditions, insurance providers, and patient age groups**. The dashboard presents key information through **KPI cards, charts, slicers, and detailed tables**, making it easier to understand healthcare data. It provides insights into **total patients, total and average billing amounts, average length of stay, admission types, insurance providers, medical conditions, and admission dates**. Overall, the project focuses on transforming healthcare data into an **interactive and easy-to-understand dashboard** for analysis and reporting.

### Objectives

- To analyse the **total number of patients**.
- To calculate the **total and average billing amount**.
- To analyse the **average length of patient stay**.
- To compare patients across different **age groups**.
- To analyse billing amounts by **insurance provider**.
- To examine patients based on different **medical conditions**.
- To provide an **interactive healthcare dashboard** using Power BI.

# KPI Cards

The dashboard uses **four KPI cards** to present the main healthcare performance indicators in a simple and easy-to-understand format. These cards provide a quick summary of the overall patient and billing information without requiring users to analyse the charts first.

# Total Patients

The **Total Patients** KPI displays **56K**, representing the total number of patients included in the healthcare dataset. This provides an overall view of the number of patient records being analysed in the dashboard.

# Total Billing Amount

The **Total Billing Amount** KPI shows **1.42bn**, representing the total billing amount recorded for all patients. This indicator provides an overview of the overall billing generated within the available healthcare data.

# Average Billing Amount

The **Average Billing Amount** KPI displays **25.54K**. It represents the average billing amount per patient and helps provide a general understanding of the typical billing value within the dataset.

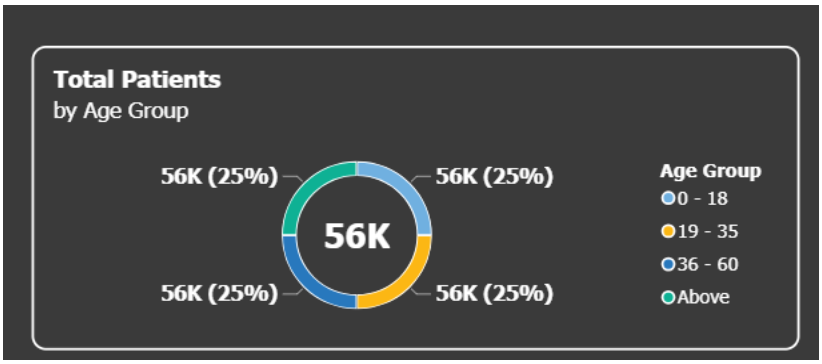
# Average Length of Stay

The **Average Length of Stay** KPI shows **15.51 days**. It represents the average number of days patients stayed in the hospital. This indicator helps provide an overview of the typical duration of patient stays.Overall, these four KPI cards provide a **quick summary of patient volume, total billing, average billing, and hospital stay duration**, forming the main overview section of the Healthcare Analytics Dashboard.



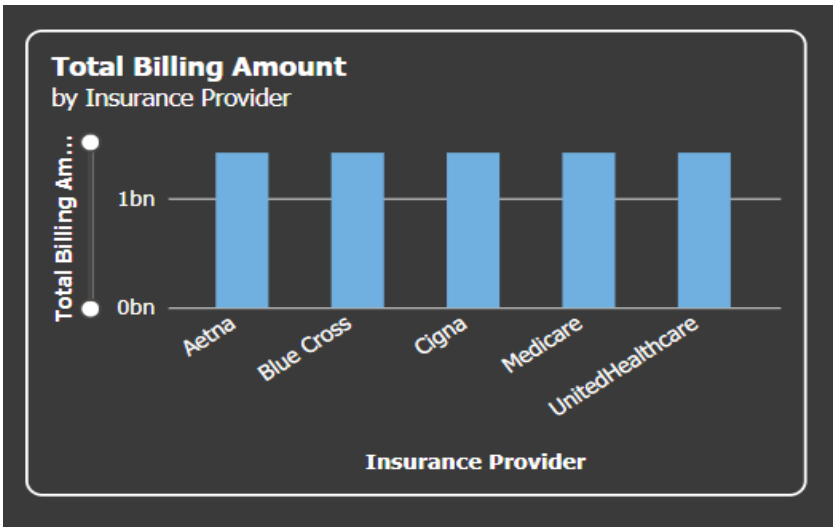
## Total Patients by Age Group

The **Total Patients by Age Group** donut chart shows the distribution of patients across four age groups: **0–18, 19–35, 36–60, and Above**. Each age group represents **25% of the total 56K patients**, showing an equal distribution across the four groups. The chart provides a quick visual comparison of patient distribution based on age and helps understand the overall age composition of the patient population.



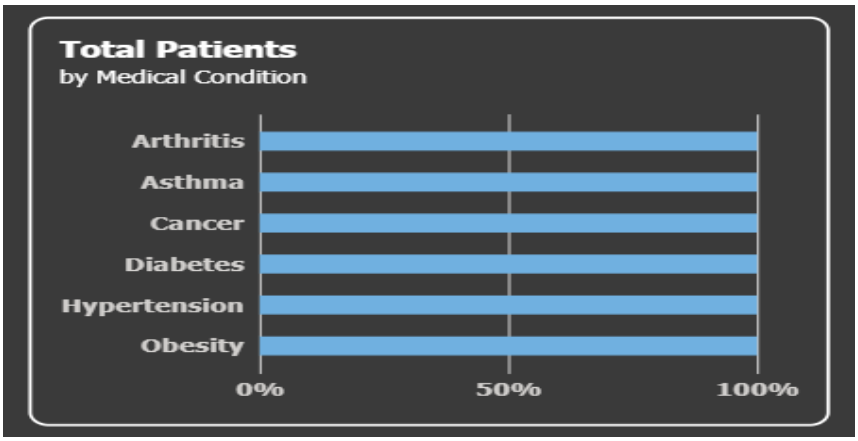

## Total Billing Amount by Insurance Provider

The **Total Billing Amount by Insurance Provider** column chart compares the billing amounts associated with different insurance providers such as **Aetna, Blue Cross, Cigna, Medicare, and UnitedHealthcare**. This helps provide an overview of billing distribution across insurance providers.



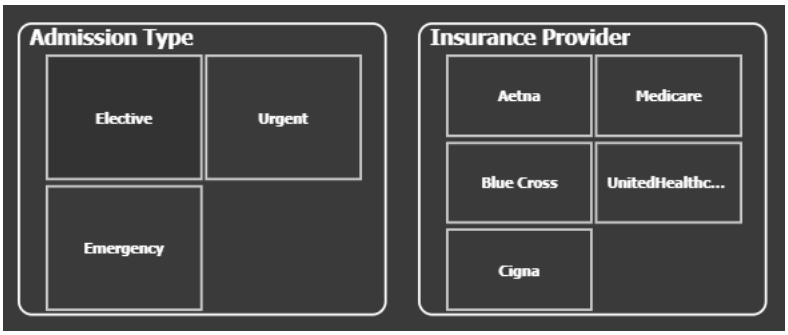
## Total Patients by Medical Condition

The **Stacked Bar Chart** is used to compare the total number of patients across different medical conditions such as **Arthritis, Asthma, Cancer, Diabetes, Hypertension, and Obesity**. The horizontal bars make it easy to compare patient counts between the conditions. The stacked bar chart provides a clear visual representation of patient distribution and makes comparison between medical conditions easier.




## Admission Type and Insurance Provider

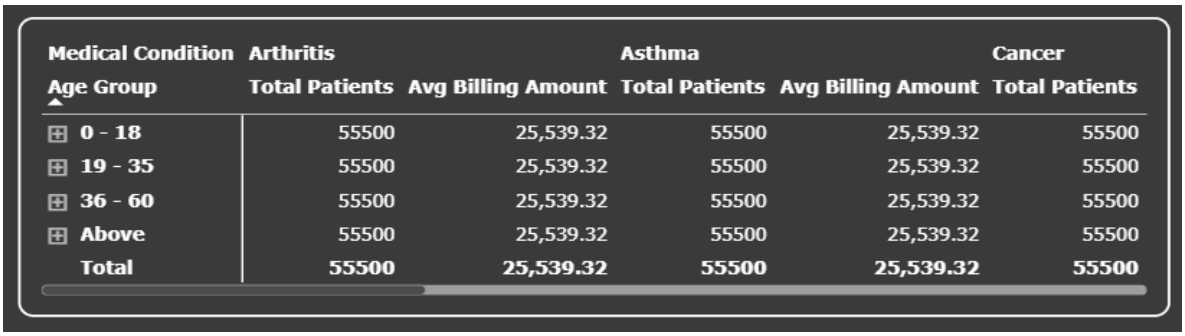
The dashboard includes **interactive slicers** for **Admission Type** and **Insurance Provider**. The admission types include **Elective, Urgent, and Emergency**, while the insurance providers include **Aetna, Blue Cross, Cigna, Medicare, and UnitedHealthcare**. These slicers allow users to select specific categories and **filter the dashboard dynamically**. This helps users analyse the corresponding **patient numbers, billing amounts, and other healthcare information** based on their selected admission type or insurance provider.



# Patient and Billing Analysis by Age, Gender and Medical Condition

A **Matrix visual** was created to provide a detailed analysis of patient and billing information. The **Rows** contain **Age Group and Gender**, while the **Columns** contain **Medical Condition**. The **Values** include **Total Patients and Average Billing Amount**.

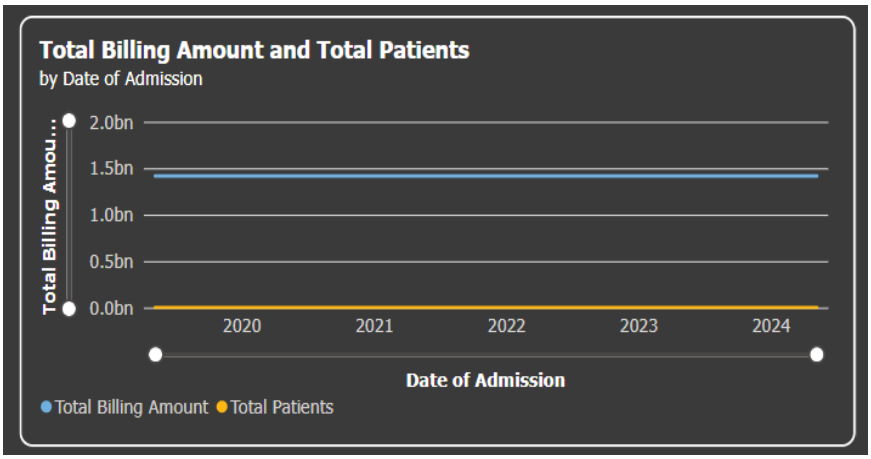
This visual helps compare patient numbers and billing amounts across different **age groups, genders, and medical conditions**. Its structured format makes it easier to view and analyse the healthcare data in detail.



| Medical Condition | Arthritis | | Asthma | | Cancer |
|---|---|---|---|---|---|
| Age Group | Total Patients | Avg Billing Amount | Total Patients | Avg Billing Amount | Total Patients |
| ⊞ 0 - 18 | 55500 | 25,539.32 | 55500 | 25,539.32 | 55500 |
| ⊞ 19 - 35 | 55500 | 25,539.32 | 55500 | 25,539.32 | 55500 |
| ⊞ 36 - 60 | 55500 | 25,539.32 | 55500 | 25,539.32 | 55500 |
| ⊞ Above | 55500 | 25,539.32 | 55500 | 25,539.32 | 55500 |
| Total | 55500 | 25,539.32 | 55500 | 25,539.32 | 55500 |

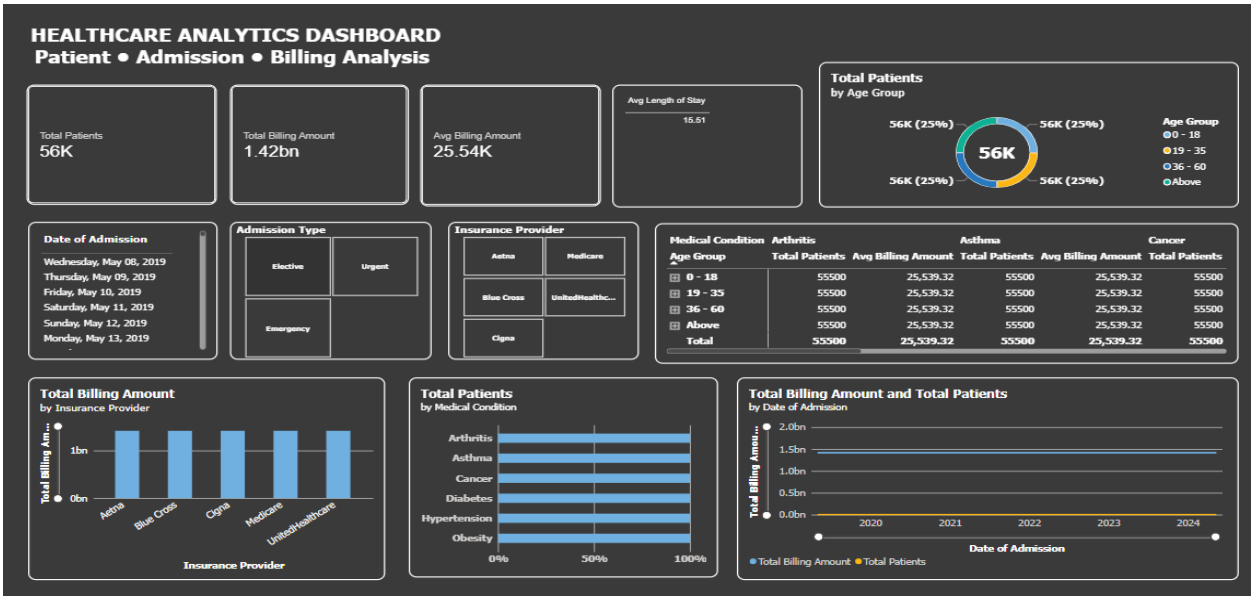
# Patient and Billing Analysis by Date of Admission

A **Line Chart** was created to display **Total Billing Amount and Total Patients by Date of Admission**. It shows patient and billing information across different admission dates, making it easy to observe changes and trends over time. The chart helps users identify variations in **patient numbers and billing amounts** and understand the overall pattern of healthcare activity during the selected period.



# Final Healthcare Analytics Dashboard

The final **Healthcare Analytics Dashboard** brings together all the important patient and billing information in a single interactive report. It includes **KPI cards, donut charts, bar charts, line charts, slicers, and a Matrix visual** to present the data clearly. The dashboard allows users to analyse **total patients, billing amounts, average billing amount, average length of stay, medical conditions, age groups, gender, admission types, insurance providers, and admission dates**. The interactive slicers help users filter the data based on their requirements, while the different visuals make it easier to identify patterns and compare healthcare information. Overall, the dashboard provides a clear and organized view of the healthcare data for effective analysis.




# Conclusion

The **Healthcare Analytics Dashboard** was successfully developed using Power BI to analyse patient and billing information. The dashboard provides clear insights into **patient numbers, billing amounts, average length of stay, medical conditions, age groups, gender, admission types, and insurance providers**. The use of KPI cards, charts, slicers, and a Matrix visual makes the data easy to understand and explore. Overall, the dashboard presents healthcare information in an **interactive, organized, and meaningful way**, supporting effective data analysis and understanding of important healthcare patterns.